

Threat Assessment Report

Executive summary for model robustness under adversarial attack

Model: 20260206_171419_resnet50_classifier.pt

Attack Type: FGSM

Images Evaluated: 9

Attack Success Rate

88.9%

Original Confidence

62.11%

Adversarial Confidence

37.73%

Execution Time

5.78s

Accuracy drop: 24.38%

Threat Level Assessment

HIGH RISK

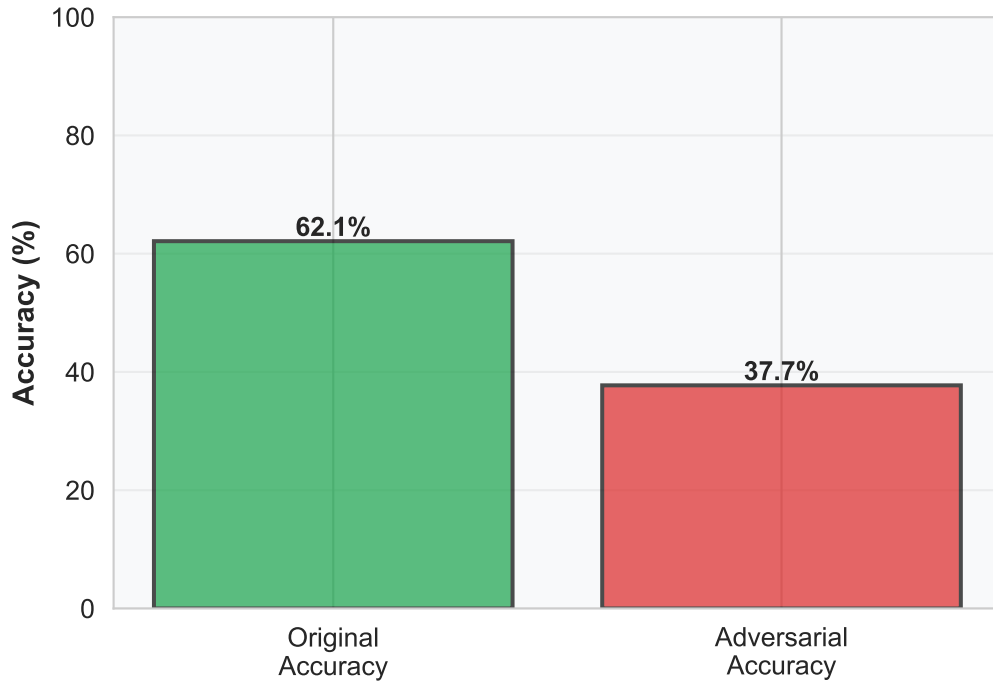
The FGSM attack altered model predictions on 88.9% of the evaluated images. Average confidence shifted from 62.11% to 37.73%, indicating the current robustness posture is high risk.

Assessment Narrative

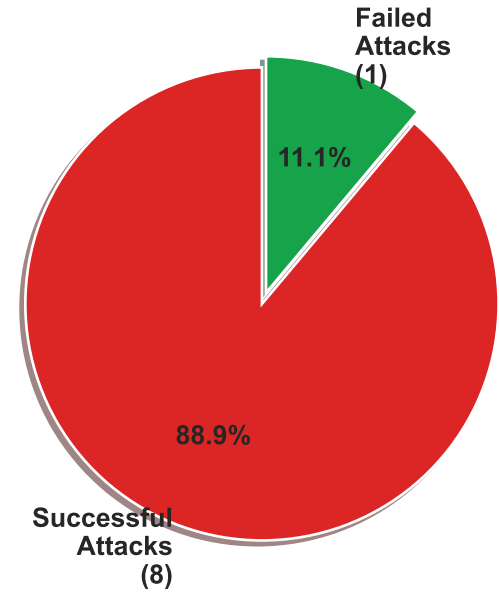
Successfully executed FGSM attack on model 20260206_171419_resnet50_classifier.pt using 9 test images. Attack success rate: 88.9%. Average original accuracy: 62.11%, Average adversarial accuracy: 37.73%. The attack successfully fooled the model in 8 out of 9 cases.

Detailed Analysis

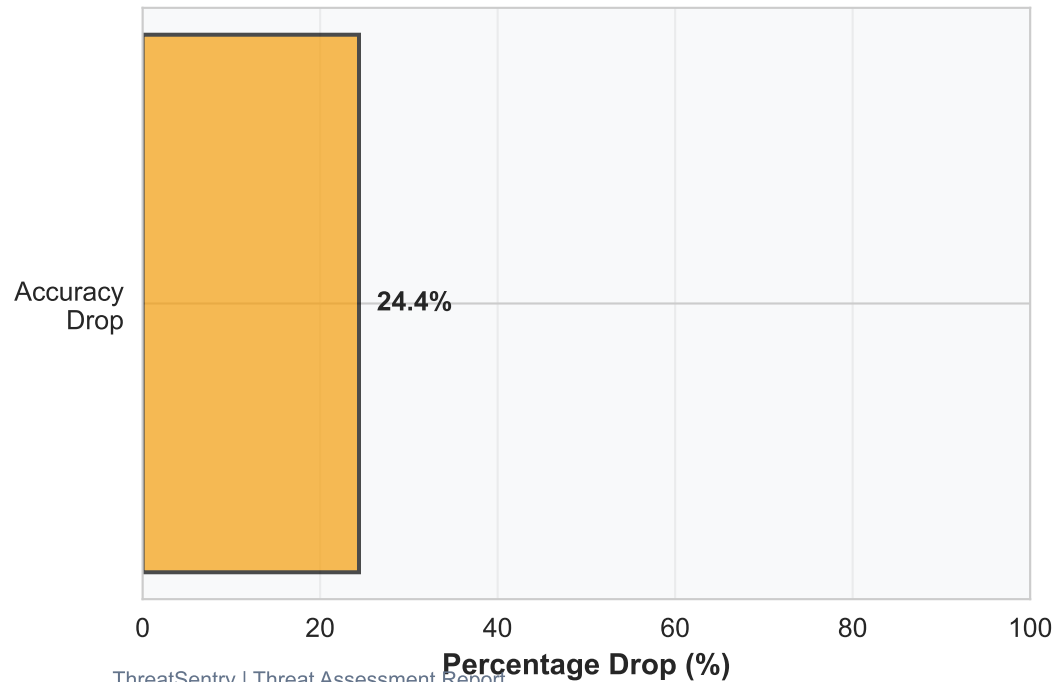
Accuracy Comparison



Attack Success Distribution



Model Robustness Impact



Confidence Comparison (Sample)

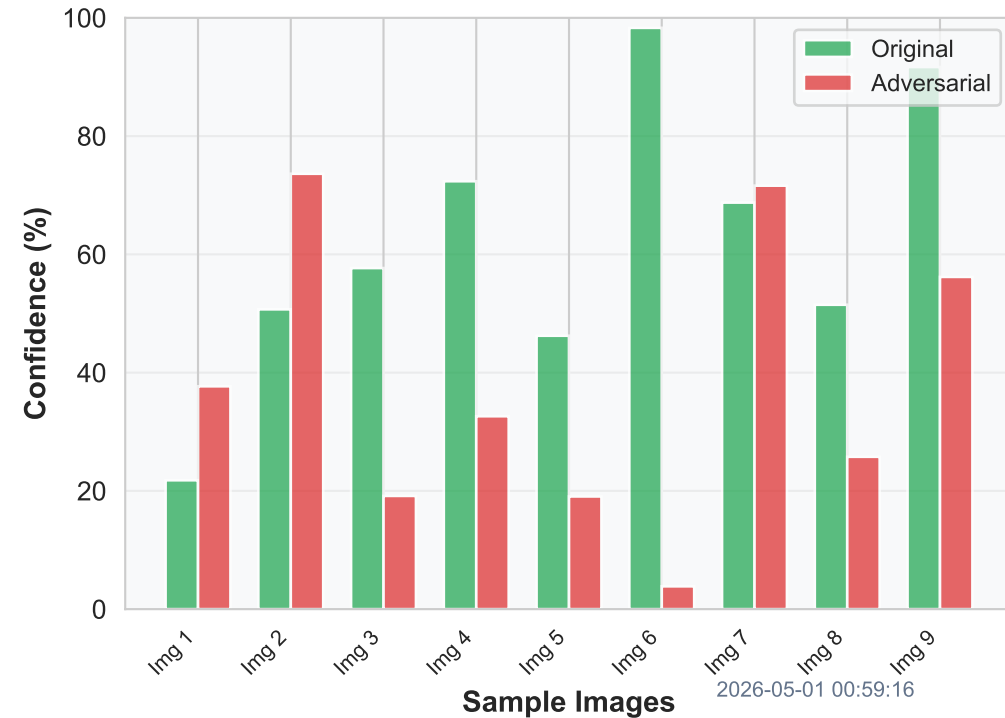


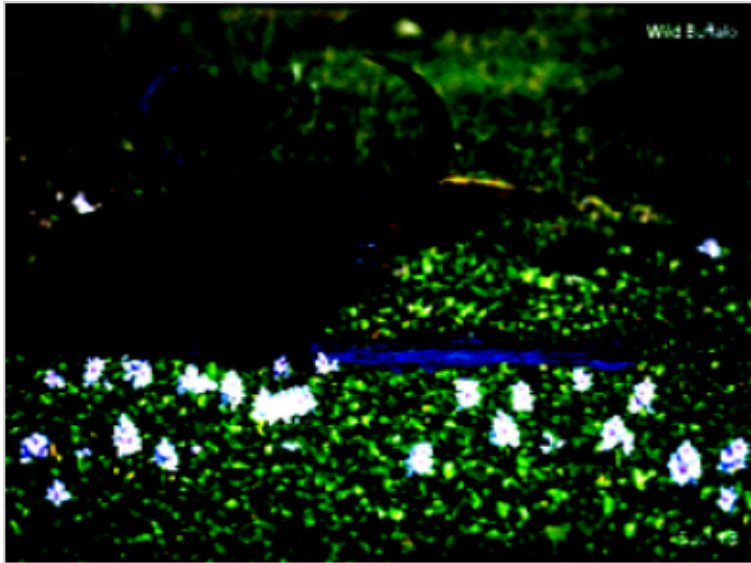
Image Evidence - Original vs Adversarial

Images 1-2 of 9

Original: Class 346
(57.7%)



Adversarial: Class 620
(19.2%)
ATTACK SUCCESSFUL



Original: Class 292
(91.7%)



Adversarial: Class 292
(56.2%)
ATTACK BLOCKED



Image Evidence - Original vs Adversarial

Images 3-4 of 9

Original: Class 335
(21.8%)



Adversarial: Class 378
(37.7%)
ATTACK SUCCESSFUL



Original: Class 346
(72.4%)



Adversarial: Class 386
(32.6%)
ATTACK SUCCESSFUL



Image Evidence - Original vs Adversarial

Images 5-6 of 9

Original: Class 101
(50.7%)



Adversarial: Class 385
(73.6%)
ATTACK SUCCESSFUL



Original: Class 274
(98.3%)



Adversarial: Class 273
(3.9%)
ATTACK SUCCESSFUL



Image Evidence - Original vs Adversarial

Images 7-8 of 9

Original: Class 353
(46.2%)



Adversarial: Class 815
(19.1%)
ATTACK SUCCESSFUL



Original: Class 272
(51.5%)



Adversarial: Class 274
(25.8%)
ATTACK SUCCESSFUL

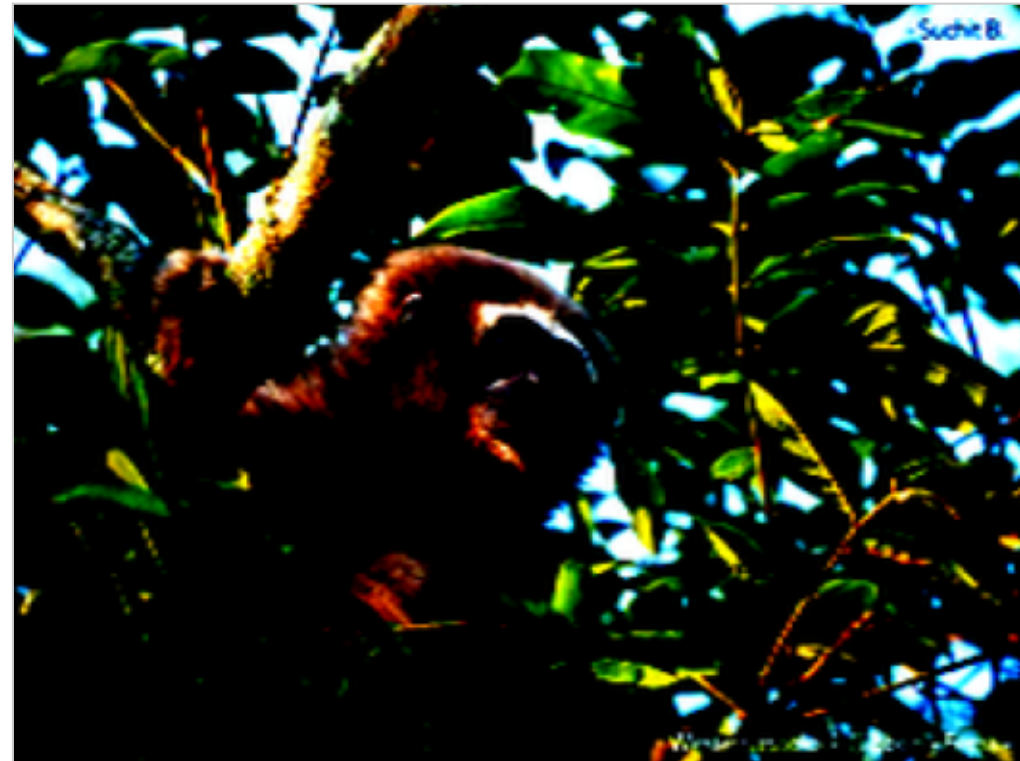


Image Evidence - Original vs Adversarial

Images 9-9 of 9

Original: Class 380
(68.8%)

Adversarial: Class 364
(71.6%)
ATTACK SUCCESSFUL



Recommendations

Security actions based on observed threat exposure

Assessment Summary

Successfully executed FGSM attack on model 20260206_171419_resnet50_classifier.pt using 9 test images. Attack success rate: 88.9%. Average original accuracy: 62.11%, Average adversarial accuracy: 37.73%. The attack successfully fooled the model in 8 out of 9 cases.

Security Recommendations

1. Implement Adversarial Training

- Retrain your model with adversarial examples to improve robustness
- Use techniques like FGSM, PGD during training phase

2. Add Input Validation & Preprocessing

- Implement input sanitization and anomaly detection
- Use defensive distillation or feature squeezing

3. Deploy Ensemble Methods

- Use multiple models with different architectures
- Implement voting mechanisms for predictions

4. Continuous Monitoring

- Set up real-time performance monitoring
- Detect and alert on unusual prediction patterns

5. Regular Security Audits

- Conduct periodic threat assessments
- Stay updated with latest attack techniques